

HYBRID SEARCH STRATEGIES

1) Combining Dense And Sparse Retrieval

Hybrid retrieval combines both dense and sparse scores (e.g. using a weighted sum or learning-to-rank methods) to improve recall and relevance.

You get the semantic power of embeddings, and the exact match of keywords — best of both worlds.

RAG

Doc ID	Content
D1	"LangChain supports agents and chains"
D2	"The Eiffel Tower is in Paris"
D3	"You can build LLM-powered apps using LangChain"

Vector store

Semantic Search

Context $\Rightarrow$ LLM
 $\Downarrow$
O/p.

Text $\rightarrow$ Sparse Matrix

NLP

Vector store

① Sparse Retrieval : Matches exact words using techniques such as TF-IDF

Query
 $\Downarrow$
Bow, TF-IDF, BM25, $\rightarrow$ Exact Search
 $\Downarrow$
Sparse Matrix Exact Keyword Search

$\Downarrow$
Exact Search

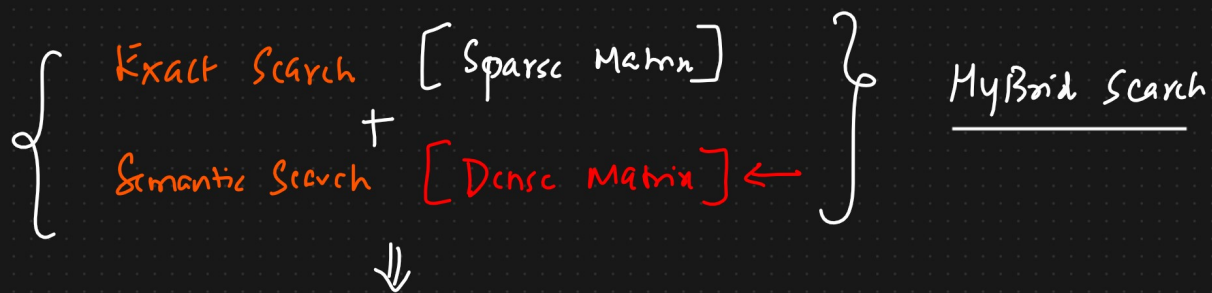
Sparse Matrix

I want to have food $\Rightarrow$ $\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

② Dense Retrieval $\Rightarrow$ Semantic Meaning Using Vector Embedding

FAISS, CHROMADB

Query $\rightarrow$ Vector store $\rightarrow$ Semantic Search
 $\Rightarrow$ O/p Similar Matching Sentences



Retrieval

- 1) Dense Retrieval (Embeddings + Cosine Similarity)
- 2) Sparse Retrieval (TF-IDF)

$$\text{Score}_{\text{hybrid}} = \alpha * \text{Score}_{\text{dense}} + (1-\alpha) * \text{Score}_{\text{sparse}}$$

$$\text{Score}_{\text{dense}} = \text{Cosine Similarity} (\text{Inp}, \text{VectorStore})$$

$$\text{Score}_{\text{sparse}} = \text{TF-IDF} (\text{Inp}, \text{Vector Representation})$$

$$\alpha = \text{Weightage} = \underline{\underline{0.5}}$$

Doc ID Text

- D1 "LangChain helps build LLM apps"
- D2 "Pinecone is used for vector search"
- D3 "Eiffel Tower is in Paris"

Your query: "build application using LLM"

Tip: Build application
using LLM

Cosine Similarity [Dense Embedding]

→

D1	0.85
D2	0.40
D3	0.10

Bow, TF-IDF, BM25

→

D1	0.60
D2	0.20
D3	0.10

⇒ Sparse Matrix.

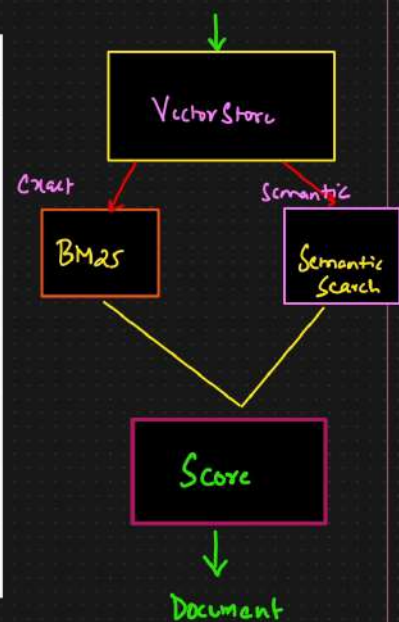
$\alpha = 0.5$

↓

$$\begin{aligned} D1 &= 0.5 \times 0.85 + 0.5 \times 0.6 = \approx 0.82 \\ D2 &= 0.5 \times 0.40 + 0.2 \times 0.5 = \approx 0.18 \\ D3 &= 0.5 \times 0.10 + 0.3 \times 0.5 = \approx 0.04 \end{aligned}$$

Benefits of Combining BM25 + Semantic Search → FAISS

Benefit	Explanation
1. Boosts Recall	BM25 may catch exact keyword matches that dense retrieval misses; dense captures meaning even if keywords differ. Together, you minimize the risk of missing relevant documents.
2. Handles Synonyms and Rephrasing	Semantic search can match "create app" with "build LLM system", even if there are no shared words. BM25 will catch exact "LLM", "app".
3. Improves Retrieval Robustness	Covers both users who search by specific terms (e.g., "Langchain agent") and those who use natural phrasing (e.g., "how do I use Langchain to talk to tools").
4. Supports Lexical Importance	BM25 scores rare keywords higher — this is crucial in technical/legal/medical contexts where a rare term (like "osteoporosis") should weigh heavily.
5. Bridges Document Diversity	In large corpora (e.g., web pages, PDFs, blogs), you often have a mix of well-structured and loosely written text. Hybrid retrieval adapts to both.
6. Easy to Tune via Weights	You can easily adjust the influence of each method: e.g., $\text{score} = 0.7 * \text{dense} + 0.3 * \text{sparse}$.
7. Helps with Misspellings	Dense models are more tolerant to typos and misspelled words (e.g., "llm aplication"). BM25 may find more. Together, you catch more cases.



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🧠 When to Use BM25 + Semantic Search Together?

🔗 Use Case

🔍 Why Hybrid Retrieval Helps

RAG Pipelines	Prevents retrieval hallucination by ensuring both exact and fuzzy matches are considered.
Technical Documentation Search	Developers may search "how to use API" while the doc says "API usage" — BM25 and semantic together improve hit rate.
Legal/Medical QA	Some queries require precise term matching (BM25), while others need general understanding (dense).
E-commerce/Product Search	"cheap noise-canceling headphones" could match "affordable ANC earbuds" — dense helps; BM25 confirms "ANC".
Multilingual or Cross-lingual Retrieval	Semantic models can bridge language differences; BM25 ensures matching if terms are in the same language.
Customer Support	Real users may type vague or keyword-heavy queries — hybrid improves retrieval reliability in chatbots/FAQs.
Transcripts/Unstructured Data	Speech data or emails may contain inconsistent phrasing — dense retrieval picks meaning, sparse supports clarity.